

Data Model: GraphDB Triples and MySQL Time-series

GraphDB (Brick/REC/ASHRAE 223)

```
@prefix brick: <https://brickschema.org/schema/Brick#>
@prefix rec: <https://w3id.org/rec#>
@prefix ashrae: <https://data.ashrae.org/standard223#>

brick:C02_Sensor_5_01 rdf:type brick:C02_Sensor .
brick:C02_Sensor_5_01 rec:isLocatedIn brick:Zone_5_01 .
brick:C02_Sensor_5_01 ashrae:hasExternalReference _:ref1 .
_:ref1 ref:hasTimeseriesId "uuid-1234" ; ref:storedAt "mysql" .
```

UUID Mapping

MySQL (wide table)

```
CREATE TABLE readings (
  ts DATETIME NOT NULL,
  uuid_1234 DOUBLE, -- C02_Sensor_5_01
  uuid_5678 DOUBLE, -- C02_Sensor_5_02
  ...
);
```

Unpivot Strategy (UNION ALL)

```
SELECT ts, 'uuid_1234' AS uuid, uuid_1234 AS val FROM readings UNION ALL .
```